

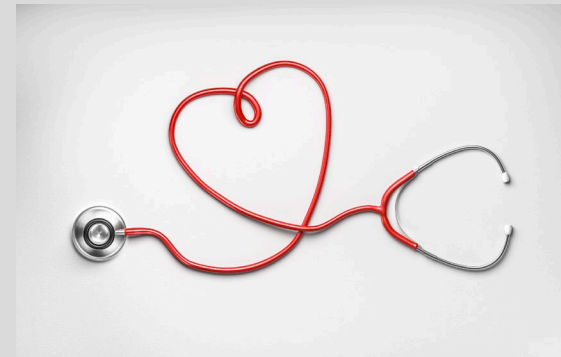
Cardiac Prevention and Improving Communication

**William Palmer M.D.
Assistant Professor
Department of Clinical Medicine
wpalmer@nyit.edu**

Session Objectives

- Introduce Prevention and cardiac risk factors found in history and physical exam
- Discuss pitfalls of communication
- Discuss methods of documenting risk factors in assessment and addressing it in the plan.
- Discuss the role of communication (written and verbal) in expressing your decision-making process including the information you are trying to communicate

Source: Course Syllabus



What is Prevention?

- “**Primary Prevention**—intervening before health effects occur”
 - Example: addressing poor diet, inactivity, tobacco use.
- “**Secondary Prevention**—screening to identify diseases in the earliest stages, before the onset of signs and symptoms”
 - Hypertension, hyperlipidemia.
- “**Tertiary Prevention**—managing disease post diagnosis to slow or stop disease progression”

Question 1

- A 55-year-old asymptomatic black male presents to the clinic.

He is concerned that his brother recently had a heart attack at age 57. He denies any symptoms of chest pain, palpitations, dyspnea, nor syncope. His father had heart disease and passed away in his 60's. He does not smoke, his BMI is 38, and his blood pressure is 140/90. After completing a complete history and physical exam, how would you end this encounter?

- A. Recommend that he lose weight.
- B. Recommend that he should eat a healthy diet.
- C. Assess other cardiac risk factors including glucose and cholesterol.
- D. Advise him that he should exercise more
- E. Advise him that it is genetics and there is nothing he can do but take medicines.

Think about what in the patient's history is cardiovascular risk factors?

Cardiovascular Risk Factors

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- First degree relatives with CVD: *found in family history*
- Tobacco and excessive alcohol use: *found in social history*
- Physical Inactivity: *found in social history*
- Poor Diet: *found in social history*
- BMI; Central Obesity: *found with vitals and exam*
- Hypertension: *found with vitals*
- Dyslipidemia: *ordered in labs*
- Diabetes/Fasting blood glucose: *ordered in labs*
- Presence of Atherosclerosis/Existing CVD: *found in diagnostics , sometimes exam.*

History: Family History

- First-degree relative with cardiovascular disease
 - **Men prior to age 55**
 - **Women prior to age 65**

ASK:

- “Does any first-degree relative have heart disease at any age?”

Document: include the **age** when they experienced heart disease/attack.

HTN: Secondary Prevention

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Blood Pressure Screening (**USPSTF**)
 - Screen ages 18-39 every 3-5 years if normal blood pressure and no risk factors for CVD.
 - Screen yearly ages 18-39 if risk factors.
 - *Remember those risk factors?*
 - Screen yearly after age 40.

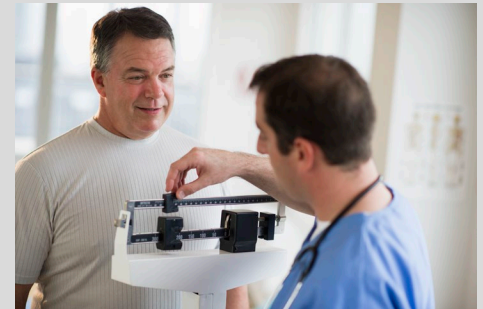
Question 2

A 46-year-old male diabetic returns to the clinic for follow up on elevated blood pressure. You have previously counselled the patient on improving their diet, increasing aerobic exercise, and weight loss. Over the last month the patient has managed to lose 5 lbs and the blood pressure has lowered from 160/100 to 135/90 today. Which is the next step?

- a. Start blood pressure medicine since diastolic is 90 and lifestyle modification failed.
- b. Continue current plan since systolic is less than 140.
- c. Evaluate and modify the current plan to better achieve goal of bp <140/90.
- d. Recommend the *Carnivore Diet*.
- e. Encourage tobacco use as it will suppress the appetite.

Let's work thru a case...

- Thomas, Danny DOB: 5/15/60
- Chief Complaint:
 - Chest Pain
- Vitals:
 - BP 140/90, HR 82, RR 18, POX 98%, BMI 26



Before you get started

- Things to consider:

What is apparent?

- Age
- Vitals
- Chief complaint

What possible risk factors could the patient have?

Tip: Prepare for your history and physical exam. Establish routine in note taking and organization of notes.

Written Documentation Purpose

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Communication:
 - In hospital help other teams understand what you are doing.
 - In clinic help you remember what you are doing.
 - At home- patient understands what is going on.
- Billing:
 - Aid insurance companies with evidence to support your efforts and to not commit Fraud.
- Legal:
 - Aid Lawyers with evidence to support how you addressed the patient's illness and concerns.

**DECISION
MAKING
PROCESS**

Case: HPI notes example

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

65yo

L Chest to Back

8/10 yard work

Sharp ↑ Breathing
moving

Tylenol X Rest +

↓ SOB, N, fever
Indigestion

Case: HPI documented

- *Danny Thomas is a 65-year-old male who reports sharp, 8/10, left chest pain that radiates to left back. It started at the end of day after doing yard work. It is worse with breathing and moving, Tylenol gave no relief, and rest helped some. He denies shortness of breath, nausea, fever, and indigestion.*

History Decision making:

- What is our differential?
 - Cardiac
 - Pulmonary
 - Musculoskeletal
- What is other valuable history information we need?
 - pmh, psx, meds, allergies, fh, social

Case: History Notes

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

AMH - HTN history
DPSX
FH M + 80 stroke
D-X82 MI
B-63-HTN PCN
DK All Rash
Cocaine
Tob - 1 ppd X30 years Yhd
Alcohol 1 beer/day
quit 15
fill. Dex, Calk - 15 alt
fast 2A

Pitfalls in Documentation

- Subjective:
 - Organization
 - Details Matter
 - Allergies
 - Social History Elements
 - ROS

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Documentation

- Organization:
 - Order: H&P / SOAP
 - Headings

Medical professionals are trained in a standardized way of documenting information. The information should be in the correct location with a standard heading.

Allergies

- Allergies:
 - Must include documentation of medicine allergies as well as environmental/food.
 - Must include reaction

Allergy documentation helps the reader be aware if there are any drug reactions they should be aware of.

Other: Latex, food, and pollen allergies could be component of presentation, concern for the medical gloves being used, and potential for cross reaction.

- *Using an EMR- always confirm*

Social history

- Social: tobacco/alcohol/drugs.
- *Documenting previous and past use helps the reader understand risk. Risk of cardiovascular disease or even cancer.*

Example:

- Patient smoked 1ppd cigarettes for 30 years but quit 15 years ago. It's great that they stopped but still has risk of cancer and CVD.

Case: History Documented

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- PMH: Hypertension
- PSX: denies
- Meds: Hydrochlorothiazide 25mg daily. (didn't take today)
- Allergies: Penicillin: rash, coconut- itching
- FH:
 - Mom deceased age 80 stroke
 - Dad deceased age 82 due to MI
 - Brother 63 alive hypertension
 - Denies children
- Social:
 - Tobacco 1ppd/30 years. Quit 15 years ago.
 - Alcohol: 1 beer/day for most of adulthood
 - Denies illicit drug use, denies formal exercise.
 - Caffeine- 1 cup coffee a day
 - Diet: high salt and fast food.

Case: ROS documented

- ROS:
- General: denies sweats, weight loss
- HEENT: denies headache, dizziness
- GI: denies indigestion, trouble swallowing.

Objective Documentation

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Objective:
 - Vitals
 - General Appearance

Vitals and General appearance helps the reader know what state the patient was in. Were they stable or unstable? In distress?

Objective Documentation

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Do not use the word “normal”
 - It is better to use a description of what “normal” is or describing things that are not there.
 - “normal” is poorly defensible.
 - Use standard terminology/abbreviations.

Case: Objective documentation

- Vitals: BP 140/90, HR 82, RR 18, POX 98%, BMI 26
- General: appears alert, awake, stated age, in no acute distress:
- Neck: supple, no tenderness, thyroid not enlarged, no lymphadenopathy. Carotids: no bruits.
- Chest wall: inhales/exhales symmetrically. tender to palpation left ribs.
- Resp: lungs clear to auscultation bilaterally. No wheezes, rales, rhonchi.
- Cardio: S1 and S2 present. PMI not displaced. No thrills. Regular rate and rhythm. No murmurs, rubs or gallops.
- OSE: L rib 5-10 inhalation somatic dysfunction

Objective

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

What about Labs?

Labs, imaging, and ecg are recorded in objective



General Lab Screening

- Cholesterol Recommendation
 - Screening in every 5 years starting at age 19.
 - Children between 9 and 11 (goal to identify familial hypercholesterolemia)

New Updates ACC/AHA

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Apolipoprotein B
 - Test once LDL-C and non-HDL-C goals are met
- Lipoprotein (a)
 - Measure at least once to identify those at higher risk.

USPSTF Prevention

- Start a Low-Moderate dose Statin for prevention of CVD mortality if ALL of the following: (*grade B*)
 - 1. Age 40-75 without history of CVD
 - 2. 1 or more CVD Risk Factors
(Dyslipidemia, **DM**, HTN or Tobacco)
 - 3. Calculated 10-year risk is greater than 10%.
- *Also may consider if above is true but with a 10-year risk 7.5%-10%. (grade C)*

ACC

- ASCVD Recommendation
- Start Statin if:
 - Presence of ASCVD
 - Diabetes: age 40-75, ldl >70, **Any Diabetic 40-75!**
 - LDL >190
 - ASCVD 10-year Risk >7.5%-9.9% consider with discussion
 - Future standard?
 - ASCVD 10-year risk 5-7.4%: low risk/consider with discussion?

Estimating Cardiac Risk

Risk Scoring Calculators:

- Newest: PREVENT ASCVD ages 30-79.
 - [PREVENT Online Calculator - Professional Heart Daily | American Heart Association](#)
 - Previous ASCVD calculator modified.
 - Removed race; added UACR, Hga1c, Zip code.
- Original ASCVD: (previous standard)
- Framingham Risk Score: (older 2008)

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Comparing Estimators

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- **ASCVD**

- Age
- Sex
- Race
- Smoking History
- Blood Pressure
- Cholesterol Levels
- History of Diabetes
- Current medications (htn, aspirin, or statin)

- **PREVENT**

- Age
- Cholesterol levels
- Blood Pressure
- BMI
- eGFR
- Also: Diabetes, smoking, medications: lipids and hypertension, zip code

Assessment

- Assessment helps the reader know what conditions you:
 - Considered (differential diagnosis)
 - Prioritized (your #1)
 - Addressed

Giving rationale helps the reader understand your decision-making process.

You should also consider risk factors in your assessment.

Modifiable Risk Factors

- Tobacco use
- Physical Activity
- Weight
- Blood Glucose
- Blood pressure
- Cholesterol
- Diet

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

*Topics
you can
counsel
on*

Case: Assessment

- DDX: Chest Pain
 1. Costochondritis: started with activity and worsened with movement. Exam elicit upon palpation.
 2. Angina: multiple risk factors: hypertension, tobacco history, family history, unknown cholesterol.
 3. Pleurisy: worse with breathing

Assessments:

1. Hypertension
2. Physical Inactivity
3. Unhealthy diet

Plan

- Plan:
 - Helps the reader understand:
 - What you did to the patient.
 - What you are going to do for the patient.
 - What you communicated to the patient along with the patient's understanding.
 - *If you ran out of time in the ICC Encounter you can still add what you would recommend in your written plan.*

Specific Diets to Recommend

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

D.A.S.H. Diet

- Dietary Approaches to Stop Hypertension
- [DASH Eating Plan | NHLBI, NIH](#)

Can you explain this diet to the patient?

Ideas: “I recommend that you should....

Increase fruits and vegetables to get to 5 servings a day,

Increase low fat or fat free dairy products,

Increase whole grain consumption,

Decrease salt intake to 1500-2300 mg a day.”

[DASH diet: Healthy eating to lower your blood pressure - Mayo Clinic](#)

Specific Diets to Recommend

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Mediterranean Diet

- <https://www.heart.org/en/healthy-living/healthy-eating/eat-smart/nutrition-basics/mediterranean-diet>

Can you explain this diet to the patient?

Ideas: *"I recommend that you should...*

Increase fruits and vegetables to get to 5 servings a day,

Increase whole grains, nuts, beans, and legumes,

Limit sugars, highly processed foods, saturated fats,

Eat more fish and poultry than red meat,

Enjoy your meal with family- not fast eating."



Plan Documentation:

- Inform/Educate/Counsel
 - Do not just document: “I counselled on diet”
 - It is not specific, informative to reader, or give an understanding of patient’s response.
 - Document: “I recommended the patient to eat a DASH diet based on her hypertension. I described the diet and the patient voiced understanding.”

Counseling Review

- [AAFP](#) {**Please Read**}
- 5 A's
- FRAMES
- BATHE
- Transtheoretical model

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Plan: Counseling Techniques

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Motivational counseling

- Patient may not be ready or willing to change.
 - *Transtheoretical Model*
- If not ready, should have patient return at future date.
 - “How about we follow up in a month and discuss this in more detail?”
- Patient given ownership of plan

Documenting Tobacco cessation:

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

I recommended the patient to quit smoking due to the increased risk it brings for heart and lung disease and cancer. Options given included medications, patches, cutting back, and counseling. Patient voiced understanding of risk and states they will work on cutting back. Follow up 1 month to see how it is going.

Counseling Example: Exercise

Physical Activity:

- Goal:
 - 90-150minutes/week aerobic exercise
 - Can be 30 minutes x5 days/week of moderate intensity.
 - Goal- raise heart rate for at least 10minutes at a time (sauna)
 - Counseling tip: lay out full recommendation, obtain feedback, if barrier, look for smaller obtainable goals with the intent to push to full goal.

OMM

- Instead of documenting:
 - “Will treat with omm”
- Consider what are you treating and how you would recommend
- Case: “Recommended treating somatic dysfunction with MET.”

Written Decision Making: Recap

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- As you follow your note from top to bottom/subjective to plan, the reader should be able to track:
 - Presentation including pertinent positives / negatives within the HPI.
 - Medical history/risk factors that could include or exclude potential for a condition.
 - Objective findings in the vitals, exam, or labs that may help confirm or deny a condition is present.
 - An assessment that connects to the subjective and objective data including differential diagnosis of the chief complaint as well as other presenting conditions.
 - A Plan that directly connects to the assessment. Make sure any recommendation/order has a direct assessment tie-in.

Verbal Communication



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Verbal

- Examples:
 - SBAR, handoffs, Oral Presentation
- Organized. Give timely information so that:
 - Listener can determine what is going on with the patient.
 - Attending can follow your decision-making process as well as form a differential diagnosis while listening.

Oral Presentation Details

- Details about the case is important
 - Including the elements of HPI
 - Including risk factors
- But not all details need to be stated.
 - Examples: non relevant.
 - History that doesn't pertain to the case.
 - However- some history that is negative is relevant to help form differential.
 - Example: lack of existing conditions, lack of risk factors.

Oral presentation

Tips:

1. Label the location/system you are examining. Don't just rattle out abbreviations:
 - Say: "eye exam- pupils equal round reactive to light."
 - Instead of saying: "P E R R L"
2. Organization is important for the listener to follow train of thought.

Case example:

Remember: Organization is key

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Give name, demographics, chief complaint, then HPI, PMH, PSX, meds, allergies, Pertinent Family history, and social risk factors.

- Example: In this case would include past history of smoking.

Give vitals, general appearance you focused exam of heart/lungs/OSE and palpation findings.

Give your differential of Costochondritis and what you recommended.

Closing